

Semantic Gesture Retrieval for real-time co-speech gestures



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Introduction

Most existing systems rely on data-driven generation, trained on either:

- **Motion capture [3]:** actor-biased and limited in variety
- **2D-to-3D estimations [4]:** cheaper but noisy due to pose tracking artifacts

These systems work, but tend to favor generic **beat gestures** and struggle to produce **semantically rich movements**.



Multiple works ([1],[2]) fill the semantic gap but lose their real-time abilities by relying on LLM retrieval for the semantic capability.



Our motivation is to propose a **gesture retrieval framework** that seeks to replace this LLM Bottleneck while ensuring:

- **Speed**, for real-time gesture selection
- **Interpretability**, by linking gestures to explicit semantic embeddings and tokens
- **Semantically aware**, by choosing gestures that fit the the context

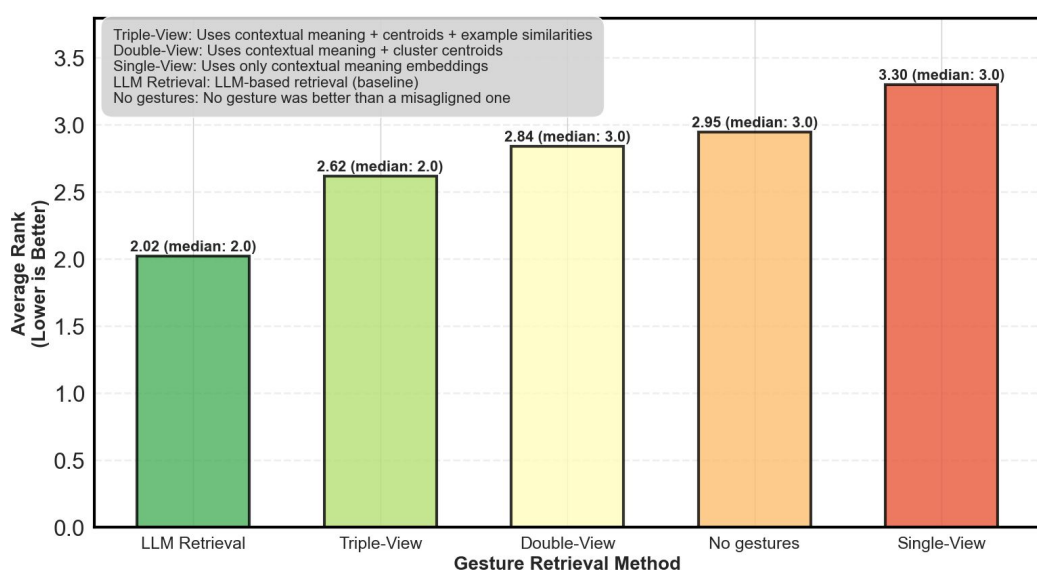
Results

We evaluated our approach using a **human-ranked dataset** of 500 synthesized utterances.

Gesture Retrieval: Compared **Single-View**, **Double-View**, **Triple-View** retrievals, **LLM Retrieval**, and **No gestures** (baseline). (information on retrieval view in the graph below)

Human Ranking: Evaluators ranked gestures by semantic relevance (1-5)

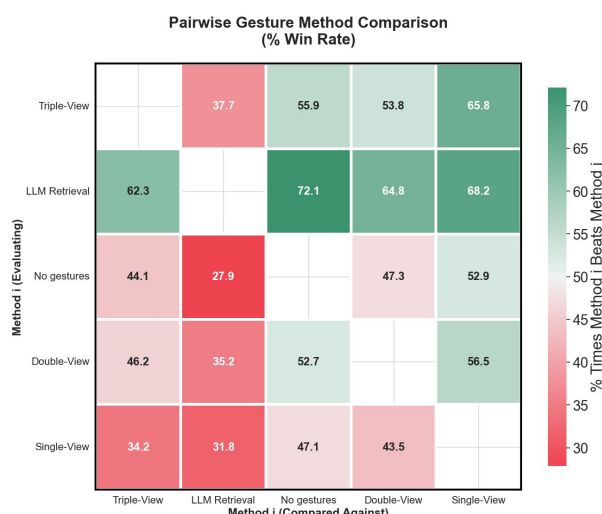
Gesture Retrieval Methods Performance Comparison (Average and Median Ranking Across All Queries)



LLM Retrieval is still the most effective method by a margin, but our Triple-View is a strong and faster alternative.

We also were able to prove the **impact of augmentation** as single view retrieval was so misaligned that “No gestures” were preferred over it on average.

The main attraction of our method comes from its **speed** and its **satisfying performance**



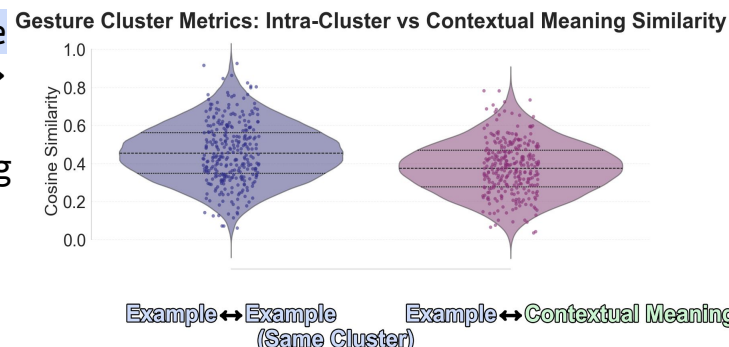
Methods

Initial SeG [1] : **Gesture Name+description** | Contextual Meaning

BELLY PAT + “The belly is patted gently with the hand” | “Appreciation”/ “Contentment”

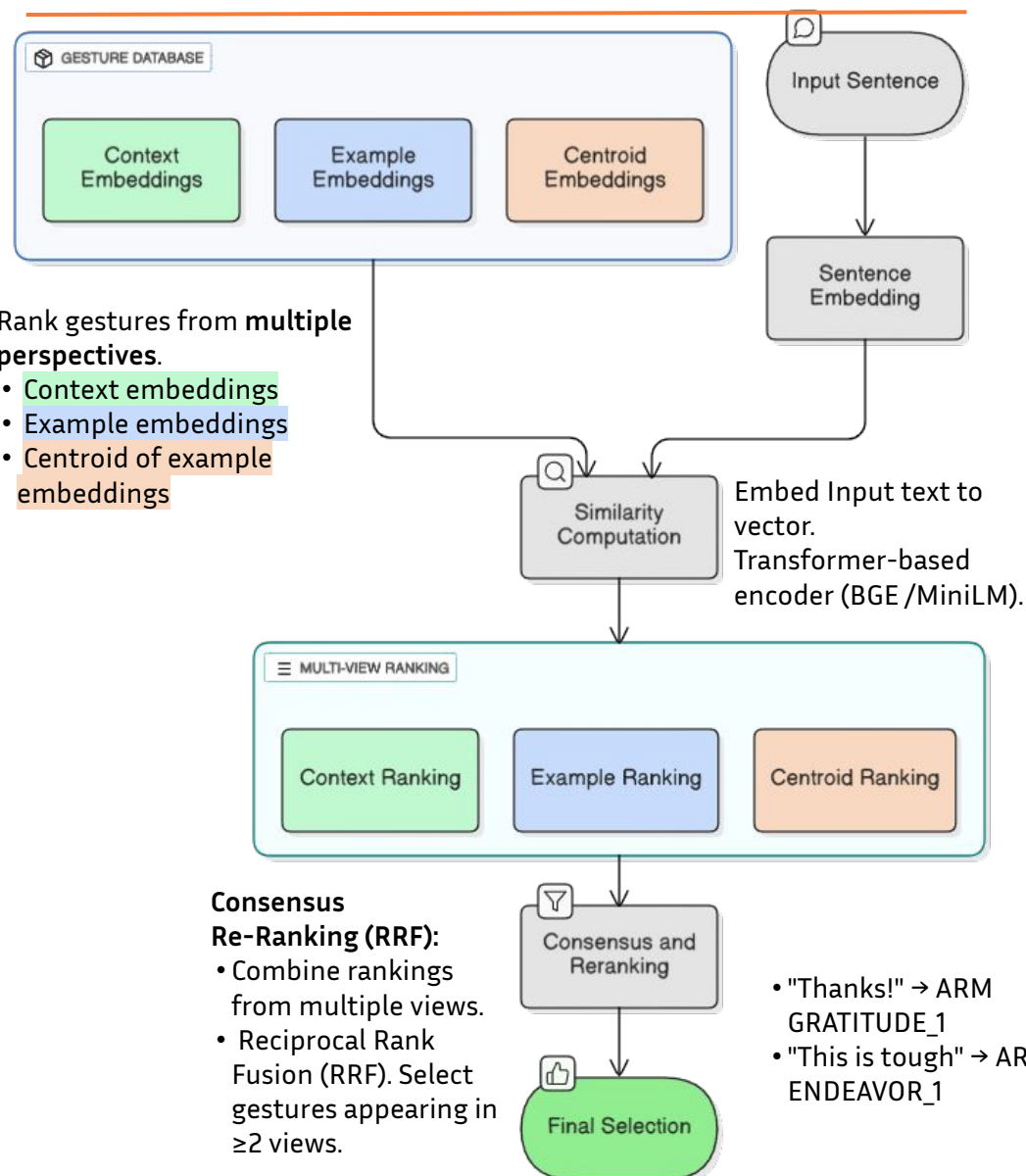
We generate 15 example sentences per gesture → broaden semantic coverage while retaining gesture meaning.

We also compute the **centroid** of these examples



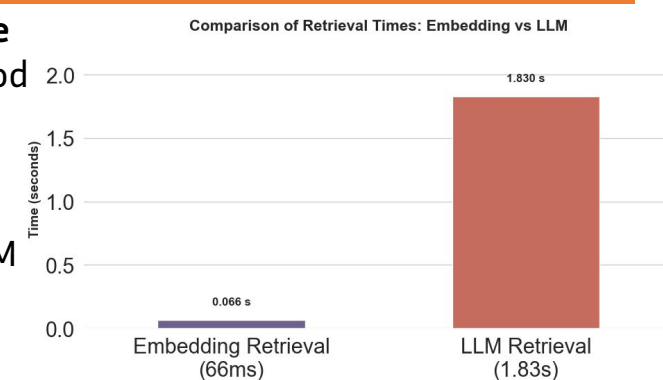
This augmentation expands the number of documents available that match the same gesture but have different vector similarities **with each other** and with **the original contextual meaning**

Augmented dataset : **Gesture** | **Contextual Meaning** | **Centroid** | **Examples**



Discussions and conclusions

This method is a **real-time simple lightweight** method to retrieve a semantic gesture for a given utterance. It is slightly inferior to LLM retrieval but **significantly faster and cheaper** !



Limitations: Lack of flexibility, dependent on semantic embedding model.

Future work: More accurate synthetic documents (examples) to also encode intent and tone and fill the rest of the gap with the LLM. Integration in data-driven frameworks